

Mime: Universal Speech, Visualized in Real-Time

COMP4107 Project Report

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Statement of Contributions

This project was completed by Daniel Lu and Methira Herath (Group 10). Both members made significant and equal contributions to nearly all aspects of the project.

Daniel was responsible for the development of the audio-to-blendshape neural model, including the architecture design, training loop implementation, and real-time inference optimization. He also was the primary contributor to the integration of the speech translation pipeline and the Zoom bridge component that allows for Mime to be run in real-time Zoom meetings.

Methira was responsible for the development of the blendshape rendering system, including the avatar rigging, mouth overlay rendering pipeline, and virtual camera integration. He also contributed significantly to model training and evaluation, as well as critical debugging and optimization work across the entire codebase.

Chapter 1

Introduction

Language barriers in video communication are no longer only a text or audio problem. Recent progress in speech-to-text, machine translation, and text-to-speech has made multilingual conversations more accessible, but live conversations over video still feel unnatural when the spoken audio and visible mouth motion do not match. This mismatch introduces an *immersion gap*: listeners hear translated speech while seeing lip movements that correspond to a different language and timing. In practical settings such as online meetings, this visual-audio inconsistency can reduce comprehension, increase cognitive load, and make interactions feel less human.

This report presents **Mime**, a real-time multimodal communication system that chains speech translation with synchronized facial animation, running on consumer hardware with latency low enough for live calls. Rather than generating synthetic face frames from scratch, Mime predicts ARKit facial blendshape coefficients directly from audio features and renders them on an animated avatar: an approach that intentionally trades some visual fidelity for the speed that *live operation* requires.

1.1 Background

Human communication is strongly multimodal. Timing, facial movement, and visual articulation all contribute to how spoken content is perceived, and disruptions to that coherence are not merely aesthetic - they measurably affect comprehension and naturalness [?]. In cross-lingual video calls, even accurate translation leaves this visual layer broken: the listener hears one language while watching lip movements shaped by another.

State-of-the-art visual dubbing systems address this problem, but they were designed for film post-production. They assume offline workflows, high-end compute, and closed pipelines - none of which are compatible with live telepresence. Mime targets the practical gap: continuous microphone capture, real-time translation, audio-driven facial animation, and direct integration with standard conferencing stacks on Windows.

1.2 Motivation and Objectives

The project is motivated by a simple goal: make multilingual video conversation feel coherent in both sound and appearance. Instead of treating translation and lip-sync as separate tools, Mime integrates them into one stream-oriented system. The platform is designed around three core objectives:

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1. **End-to-end integration:** Build a real-time pipeline that chains speech recognition, machine translation, and synthesized speech with an audio-to-blendshape visual model. This project will focus on specifically **English-to-French translation**.
 2. **Temporal visual coherence:** Produce stable lip-sync by mapping speech features to 52 ARKit facial blendshape coefficients and rendering an animated avatar suitable for live mouth overlay on a virtual camera output.
 3. **Deployment practicality:** Maintain low latency on consumer-grade hardware, with integration support for common video conferencing workflows. This project will target **Zoom** as the primary use case, but the architecture should be adaptable to other platforms with the use of virtual camera and audio interfaces.

In implementation terms, Mime uses modular processors for STT, MT, and TTS, then routes synthesized audio to both playback and a fast inference path for facial animation. This architecture enables synchronized media output and reduces duplicated processing in live operation.

1.3 Related Prior Work

The main tools people have built for this are visual dubbing systems. Wav2Lip [?] masks the lower face of a video and inpaints it conditioned on audio, achieving lip sync close to real recordings. VideoReTalking [?] adds expression normalization and face enhancement on top of a similar approach, producing cleaner results. SadTalker [?] goes further, predicting 3D morphable model coefficients from audio to get more natural head movement than purely pixel-level methods. All three produce good output, but they assume pre-recorded video and offline compute. None are designed for a live call.

The parametric side of the literature is more relevant to Mime’s approach. VOCA [?] showed you can map audio features directly to a structured face representation, specifically 3D mesh vertex displacements, and animate a wide range of identities from a single model. Mime follows the same principle but targets ARKit blendshape coefficients, which are lighter to infer and render directly on avatar-based output. For the speech side, Whisper [?] provides a practical multilingual transcription foundation, trained on 680,000 hours of audio across dozens of languages with near-human accuracy in a zero-shot setting.

The gap none of these works address is the *full pipeline*: translation, synthesis, and synchronized facial animation running together in real time. That is the specific scope Mime targets.

Chapter 2

Methods

2.1 Methods from Neural Networks

Mime is implemented as two coupled neural pipelines: a **speech translation pipeline** (STT $\rightarrow$ MT $\rightarrow$ TTS) and an **audio-to-blendshape (ABS) pipeline** that drives a 3D ARKit avatar in real time.

2.1.1 Speech Translation Pipeline

Speech-to-Text (STT): The microphone stream is captured at 16 kHz mono and segmented online using an RMS-based voice activity strategy. Chunks are finalized after a silence window and transcribed with Groq’s `whisper-large-v3-turbo`. This chunked design reduces unnecessary API calls and keeps inference focused on complete utterances, rather than segments of non-contextual speech.

Machine Translation (MT): Each English utterance is translated to conversational French using `llama-3.3-70b-versatile` through a constrained system prompt. A short rolling context buffer is included to improve coherence for fragments and pronoun resolution during live dialogue.

Text-to-Speech (TTS): French text is synthesized by Inworld TTS (default model `inworld-tts-1.5-mini`, French voice). Audio is streamed as LINEAR16[?] at 48 kHz so playback can begin before sentence completion, which lowers perceived latency in meetings.

2.1.2 Audio-to-Blendshape Neural Model

The ABS model maps raw 16 kHz mono audio to 52 ARKit blendshape coefficients at 60 FPS. It follows an encode-then-predict structure: a four-layer progressive 1D convolutional backbone extracts hierarchical phonetic features with increasing channel depth, a Transformer encoder models long-range co-articulation dependencies, and a two-layer regression head with a final sigmoid activation constrains outputs to $[0, 1]$. The model supports two backbone modes: a custom CNN (Mode B, used in all experiments) and an optional pretrained Wav2Vec2 encoder [?] (Mode A), included for future comparison. Table ?? summarises the custom CNN configuration.

Loss. The primary objective is mean squared error over predicted and target blendshape sequences:

$$\mathcal{L}_{\text{mse}} = \frac{1}{N} \sum_t \|\hat{Y}_t - Y_t\|_2^2$$

Table 2.1: ABS architecture ($H=512$, $L=4$, heads= 8).

Layer	Key Attributes	Out Dim	Params
Conv1 + BN + ReLU	$C=64$, kernel=10, stride=4	$T/4 \times 64$	0.8 K
Conv2 + ReLU	$C=128$, kernel=4, stride=4	$T/16 \times 128$	32.9 K
Conv3 + ReLU	$C=256$, kernel=4, stride=4	$T/64 \times 256$	131.3 K
Conv4 + ReLU	$C=H$, kernel=4, stride=4	$T/256 \times H$	524.8 K
Transformer Encoder	L layers, $d=H$, heads=8, FFN dim=2048	$T/256 \times H$	12.6 M
Regressor Head	Linear($H \rightarrow 256$) + LeakyReLU(0.2) + Linear($256 \rightarrow 52$) + Sigmoid	$T/256 \times 52$	144.7 K
Total			13.4 M

Early experiments produced jittery animation, so a velocity consistency term was added to penalise erratic frame-to-frame changes:

$$\mathcal{L} = \mathcal{L}_{\text{mse}} + \lambda \mathcal{L}_{\text{vel}}, \quad \mathcal{L}_{\text{vel}} = \frac{1}{N'} \sum_t \|(\hat{Y}_t - \hat{Y}_{t-1}) - (Y_t - Y_{t-1})\|_2^2$$

$\lambda=0.5$ was fixed after informal testing balanced smoothness against per-frame MSE. Training uses the Adam optimizer with gradient clipping (max norm = 1.0).

Hyperparameter Search. A grid search was run over the parameters most likely to affect capacity and stability, shown in Table ?? . The best configuration was selected by lowest validation MSE; $H=512$ consistently outperformed $H=256$, and longer training (20 epochs) improved convergence where early stopping was not triggered.

Table 2.2: Hyperparameter search space and selected values.

Hyperparameter	Values Tried	Best
Learning rate	1×10^{-4} , 5×10^{-5}	1×10^{-4}
Hidden dim H	256, 512	512
Batch size	4	4
Epochs	5, 20	20
Pretrained backbone	False	False

2.1.3 Real-Time Inference and Rendering

At runtime, the ABS model runs on a sliding audio window (0.4 s at 48 kHz) with queue-based frame updates. Predicted blendshapes are remapped from BEAT ordering to canonical ARKit ordering and applied to morph targets in a rigged GLB avatar via an offscreen renderer. A transport bridge publishes rendered frames to a virtual camera and TTS audio to a virtual cable device, enabling direct integration with Zoom-compatible workflows.

2.2 Dataset and Data Processing

The model is trained on the **BEAT multimodal dataset** [?] (ECCV 2022), a 76-hour motion-capture corpus from 30 speakers annotated with synchronized audio, text, emotion labels, and *52-dimensional facial blendshape weights* - matching the ARKit target space exactly. The English subset (`beat_english_v0.2.1`, 60 h) is used, though additional languages available in BEAT could be explored in future work.

Each sample pairs a `.wav` and `.json` file: audio is loaded as mono at 16 kHz and clipped to 10 seconds, with blendshape targets aligned at 60 FPS consistent with BEAT’s native capture rate. When a clip is truncated, the target frame count is scaled proportionally to maintain audio-target alignment. Variable-length sequences are padded per batch. An 80/20 train/validation split with a fixed random seed ensures reproducibility.

2.3 Validation Strategy and Experiments

2.3.1 Model-Level Validation

Validation runs every epoch using MSE and MAE over the held-out 20% split. TensorBoard logs train and validation loss curves alongside MAE per epoch, and best checkpoints are saved by lowest validation MSE. To assess the contribution of the velocity consistency term, models trained with $\lambda = 0.5$ are compared against MSE-only runs ($\lambda = 0$), with MAE used to measure any improvement in per-coefficient smoothness.

2.3.2 Hyperparameter Experiments

Grid search trials are run over learning rate, hidden dimension, and epoch budget (Table ??). Each trial writes isolated TensorBoard logs and saves the best checkpoint for direct comparison of convergence behavior and final validation performance.

2.3.3 Iterative/Experimental Model Development

Live testing of the initial model revealed a consistent temporal synchronisation failure: predicted blendshapes would rush ahead of the speech signal, producing visibly misaligned lip motion. This was attributed to the absence of positional encoding and no explicit residual temporal mixing, limiting the model’s ability to resolve fine-grained frame ordering. Late in the project, a revised architecture was trained but unfortunately failed to meet the speed constraints of the project, so it is unused in the final implementation. Key changes include: a three-layer convolutional backbone (strides 2-2-2, SiLU activations, BatchNorm throughout); a depthwise-separable **temporal mixer** with a residual connection; **sinusoidal positional encoding** with pre-Transformer LayerNorm; and a Transformer encoder with `norm_first=True` and GELU activation for improved training stability.

2.3.4 System-Level Validation

The full pipeline is validated in end-to-end live runs: microphone speech → transcription → translation → French synthesis → synchronized avatar mouth overlay via Zoom bridge. Concrete pass criteria are defined: first-audio-chunk latency below 800 ms; translation quality verified by manual review of 20 sampled utterances; and lip-sync coherence maintained with no perceptible frame drops over a sustained 5-minute session. Robustness under device reconnect and stream interruption is additionally assessed from operational logs.

Chapter 3

Results

3.1 Quantitative Data

All of the following results can be found in the results folder of the project GitHub repository, if too small to view here: <https://github.com/2026W-COMP4107/mime/tree/main/results>.

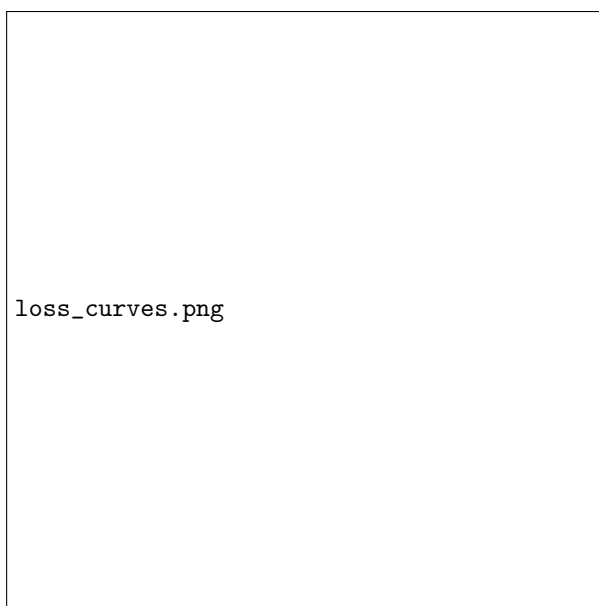
3.1.1 Hyperparameter Search Space

A large part of our pipeline in making the model look accurate is reliant on the optimization of the Audio-to-Blendshape neural network. In order to find the best hyperparameters for this model, we performed a grid search across a defined hyperparameter space. Pre-trained weights were disabled for all trials to isolate the learning capacity of the architecture from scratch. The search space consisted of 8 distinct configurations, varying the following discrete hyperparameters while maintaining a constant batch size of 4:

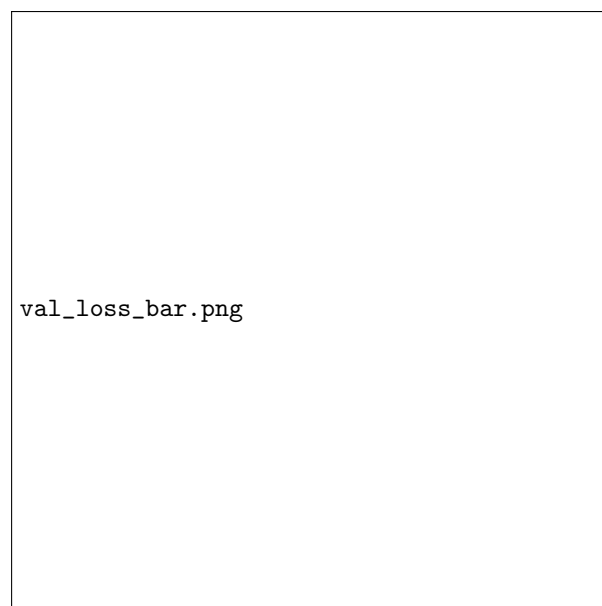
- **Epochs:** 5, 20
- **Hidden Dimensions:** 256, 512
- **Learning Rate (LR):** 0.0001, 0.00005

The performance of each configuration evaluated in the above figures used Mean Squared Error (MSE), allowing us to characterize how well each model generalized to unseen data. The quantitative results revealed several distinct trends across the hyperparameter space. Figure ?? illustrates the combined training and validation loss curves for all configurations, showing a consistent decrease in loss over time, with a more pronounced drop after the third epoch. This indicates that the model was able to learn effectively from the training data, with diminishing returns as training progressed. It also should be noted that the validation loss closely follows the training loss, suggesting that the model is not overfitting to the training data, at least within the epoch range tested.

The most significant divergence in model performance was dictated by the number of epochs as seen in Figure ?. Models trained for 20 epochs consistently outperformed those trained for only 5 epochs by a visible margin, whereas the average validation MSE across hidden dimensions and learning rates were minimal. This suggests that the model's capacity to learn from the data was more heavily influenced by the duration of training rather than the specific architecture or learning rate, within the ranges tested. This can also be seen in figure ?, where both the training and validation loss curves drop significantly after 3 timesteps. Note that the bars in Green indicate a better average peak validation loss compared to the other



(a) Combined training and validation loss curves over time.



(b) Average validation loss across all hyperparameter configurations.

Figure 3.1: Quantitative results of the hyperparameter grid search identifying the optimal model configuration.

configuration for every specific hyperparameter. Coincidentally, the model with the best peak validation loss was trained over 20 epochs, with a batch size of 4, a hidden dimension of 512, and a learning rate of 0.0001, achieving a validation MSE of 0.0096, which is a significant improvement over the worst performing model with a validation MSE of 0.0116. The best performing model achieved a peak MSE of 0.0098 on the held-out test set, which again indicates the model's ability to generalize to unseen data.

3.2 Qualitative Results

Chapter 4

Discussion

4.1 Limitations and Challenges

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4.2 Directions for Future Work

There are several concrete directions to improve Mime into a production-ready system. First, end-to-end latency should be reduced further through tighter streaming orchestration, smaller translation/synthesis models, and improved buffering policies between STT, MT, TTS, and avatar inference. Lower and more stable latency would make turn-taking feel more natural in real conversations.

Second, visual quality can be improved by increasing temporal stability of predicted blendshapes and adding stronger regularization against jitter. Future training should expand beyond the current setup with broader multilingual and speaker-diverse data, stronger validation on held-out speakers, and ablations on architecture and loss terms.

Third, the platform should evolve from a prototype integration into a robust user-facing tool. This includes better device handling, clearer fallback behavior when external APIs fail, configurable quality-latency presets, and simpler setup for common conferencing environments. Together, these changes would substantially improve reliability and usability in day-to-day deployment.

4.3 Implications and Impact

Mime’s primary impact is in improving cross-lingual communication quality by addressing both audio and visual coherence, not translation alone. By coupling translated speech with synchronized facial animation, the system targets a common failure mode in multilingual calls: hearing one language while seeing mouth motion from another. Reducing this mismatch can improve perceived naturalness and lower cognitive load for participants.

From a practical perspective, Mime demonstrates that real-time multimodal translation interfaces can run on consumer hardware using modular, API-assisted components. This lowers the barrier for educational teams, small organizations, and independent developers to prototype advanced communication tools without large infrastructure budgets.

More broadly, the project provides a reusable systems template for future telepresence applications where timing alignment between modalities is critical, including accessibility support, multilingual tutoring, and remote collaboration.

4.4 Conclusions

This project delivers a functional proof-of-concept for Mime: an end-to-end real-time pipeline that combines speech recognition, machine translation, speech synthesis, and audio-driven 3D facial animation. The implementation validates the central thesis of this work: live multilingual communication benefits when translation and lip-sync are treated as a single integrated problem.

Although limitations remain in latency robustness, model generalization, and production hardening, the current system establishes a strong technical baseline and demonstrates feasibility on commodity hardware. With continued optimization and broader evaluation, Mime can mature into a practical tool for more natural and immersive cross-language video communication.

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